

mb-pc40s-2018-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

(4nli

A group of 7 friends decide to go to a movie.

Determine how many ways the friends can sit in a row if two of the friends refuse to sit next to each other.

Solution

$$7! - 6!2! = 3600 \text{ ways}$$

$\frac{1}{2}$ mark for $7!$

1 mark for product of $6!2!$

($\frac{1}{2}$ mark for $6!$, $\frac{1}{2}$ mark for $2!$)

$\frac{1}{2}$ mark for subtraction

2 marks

Question 2

(inli)

Gabrielle listens to her radio at a sound level of 80 dB . She attended a music concert that had a sound level of 115 dB . Determine how many times more intense the music concert was than the radio.

You may use the formula:

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

where β is the intensity level of sound, measured in dB

I is the intensity of sound

I_0 is the standard minimum intensity that a person can hear

Solution

Radio:

$$80 = 10 \log \left(\frac{I}{I_0} \right)$$

$$8 = \log \left(\frac{I}{I_0} \right)$$

$$10^8 = \frac{I}{I_0}$$

$\frac{1}{2}$ mark for exponential form

Music concert:

$$115 = 10 \log \left(\frac{I}{I_0} \right)$$

$$11.5 = \log \left(\frac{I}{I_0} \right)$$

$$10^{11.5} = \frac{I}{I_0}$$

1/2 mark for exponential form

$$\begin{aligned} \frac{\text{intensity of music concert}}{\text{intensity of radio}} &= \frac{10^{11.5} I_0}{10^8 I_0} \\ &= 10^{3.5} \\ &= 3162.27766 \\ &= 3162.278 \end{aligned}$$

1 mark for comparison

2 marks

Question 3

Solve, algebraically.

$$2(7)^x = 3^{2x-3}$$

Solution

$$\log(2(7^x)) = \log 3^{2x-3}$$

$$\log 2 + x \log 7 = (2x - 3) \log 3$$

$$\log 2 + x \log 7 = 2x \log 3 - 3 \log 3$$

$$\log 2 + 3 \log 3 = 2x \log 3 - x \log 7$$

$$\log 2 + 3 \log 3 = x(2 \log 3 - \log 7)$$

$$\frac{\log 2 + 3 \log 3}{2 \log 3 - \log 7} = x$$

$$15.872483 = x$$

$$15.872 = x$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for product law

1 mark for power law

$\frac{1}{2}$ mark for collecting terms with x

$\frac{1}{2}$ mark for isolating x

$\frac{1}{2}$ mark for evaluating quotient of logarithms

4 marks

Question 4

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta + 2 \csc \theta - 8 = 0$$

Solution

$$(\csc \theta + 4)(\csc \theta - 2) = 0$$

$$\csc \theta = -4 \quad \csc \theta = 2$$

$$\sin \theta = -\frac{1}{4} \quad \sin \theta = \frac{1}{2}$$

$$\theta_r = 0.252680$$

$$\theta = 3.394 \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = 6.031$$

1 mark for solving for $\csc \theta$

1 mark for reciprocal

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

$$\theta = 3.394 \quad \theta = 0.524$$

$$\theta = 6.031 \quad \theta = 2.618$$

Question 5

You have forgotten the code to unlock your cell phone. You know the code is made up of four numbers from 0 to 9 .

Determine the number of possible codes, if repetition is allowed.

Note: A calculator is not required for the remaining test questions.

Solution

$$10 \cdot 10 \cdot 10 \cdot 10 = 10000$$

1 mark

Question 6

In the binomial expansion of $\left(\frac{7}{x^3} - 3x^7\right)^n$, the 5th term contains x^7 .

Determine the value of n .

Solution

$$x^7 = \left(\frac{1}{x^3}\right)^{n-4} (x^7)^4$$

$$x^7 = (x^{-3})^{n-4} (x^7)^4$$

$$x^7 = x^{-3n+12+28}$$

$$7 = -3n + 40$$

$$-33 = -3n$$

$$11 = n$$

1 mark for $k = 4$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for solving for n

2 marks

Question 7

Given the domain of $f(x)$ is $\{-6, 1, 3, 4\}$ and the range of $f(x)$ is $\{-4, 7, 10, 15\}$, state the domain of $f^{-1}(x)$.

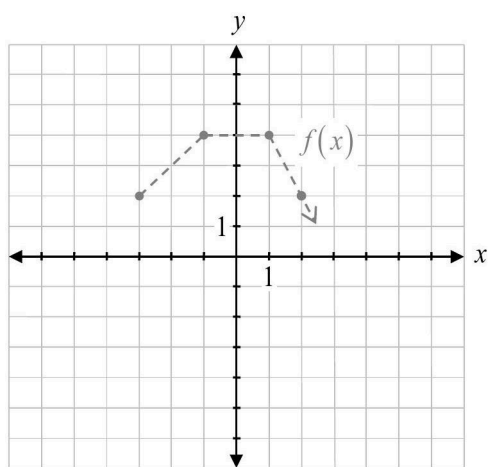
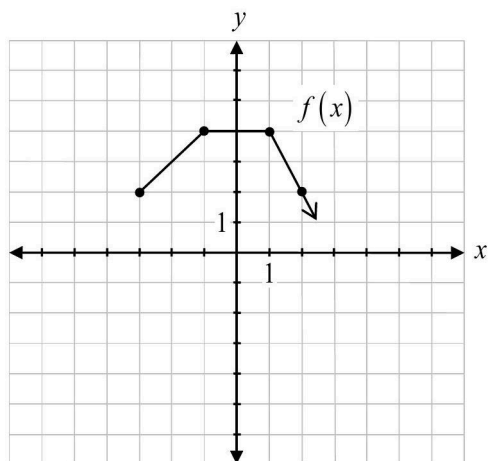
Solution

$$\{-4, 7, 10, 15\}$$

1 mark

Question 8

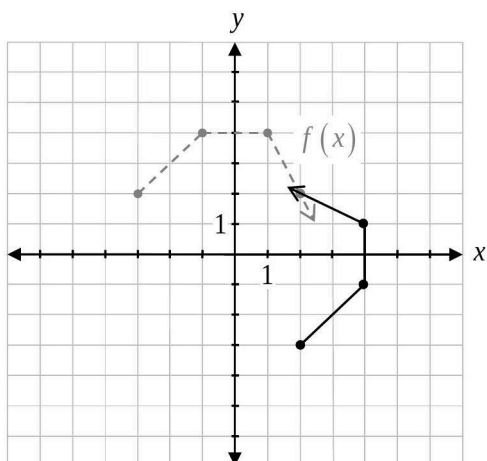
Given the graph of $y = f(x)$, sketch the graph of its inverse.



The graph of
 $f(x)$ has already
been drawn for
your reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution



Question 9

Prove the following identity for all permissible values of θ .

$$\frac{1 + \cos \theta}{1 - \sin^2 \theta} = \sec \theta + \tan^2 \theta + 1$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

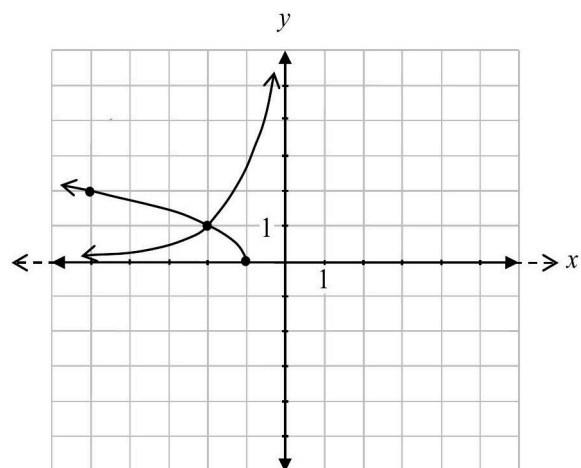
Left-Hand Side	Right-Hand Side
$\frac{1 + \cos \theta}{1 - \sin^2 \theta}$ $\frac{1 + \cos \theta}{\cos^2 \theta}$ $\frac{1}{\cos^2 \theta} + \frac{1}{\cos \theta}$ $\sec^2 \theta + \sec \theta$ $\tan^2 \theta + 1 + \sec \theta$	$\sec \theta + \tan^2 \theta + 1$ 1 mark for algebraic strategies 1 mark for logical process to prove the identity 1 mark for correct substitution of appropriate identities 3 marks

Method 2

Left-Hand Side	Right-Hand Side	
$\frac{1 + \cos \theta}{\cos^2 \theta}$	$\frac{1}{\cos \theta} + \sec^2 \theta$ $\frac{1}{\cos \theta} + \frac{1}{\cos^2 \theta}$ $\frac{\cos \theta + 1}{\cos^2 \theta}$	1 mark for algebraic strategies 1 mark for logical process to prove the identity 1 mark for correct substitution of appropriate identities 3 marks

Question 10

Thomas used graphs to solve the equation $e^{x+2} = \sqrt{-(x+1)}$.



He incorrectly states the solution as $(-2, 1)$.

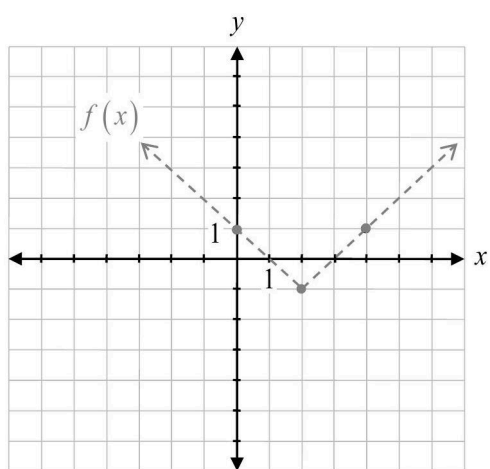
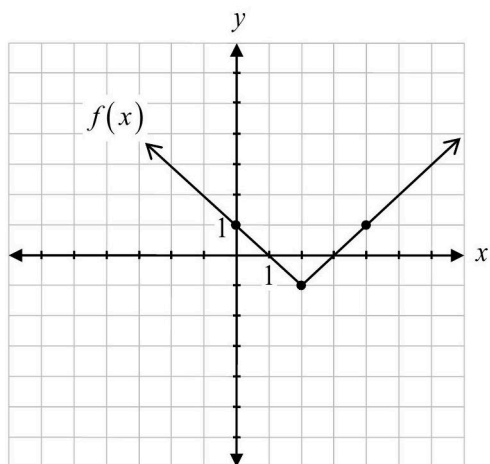
Describe how Thomas should have stated the solution.

Solution

He stated his solution as a coordinate point; his solution should have only been the value of x .

Question 11

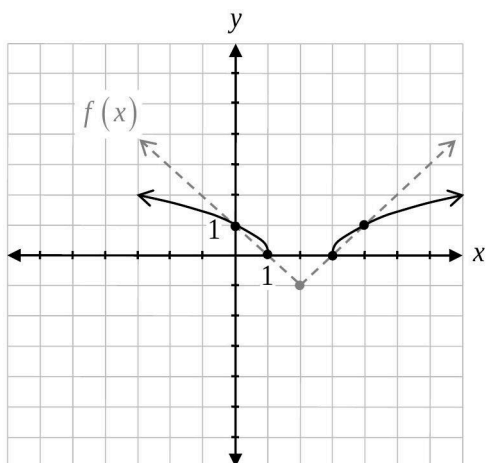
Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of
 $f(x)$ has already
been drawn for
your reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between both pairs of
invariant points

$\frac{1}{2}$ mark for shape above both pairs of
invariant points

2 marks

Question 12

When a polynomial, $P(x)$, is divided by $(x - 2)$ the resulting equation is

$$\frac{P(x)}{x-2} = x^2 - x + 1 + \frac{3}{x-2}.$$

a) Explain why $x - 2$ is not a factor of $P(x)$.

b) Determine the equation for the polynomial function $P(x)$.

$$P(x) =$$

Solution

a) There is a remainder when $P(x)$ is divided by $x - 2$.

1 mark

b) $P(x) = (x - 2)(x^2 - x + 1) + 3$

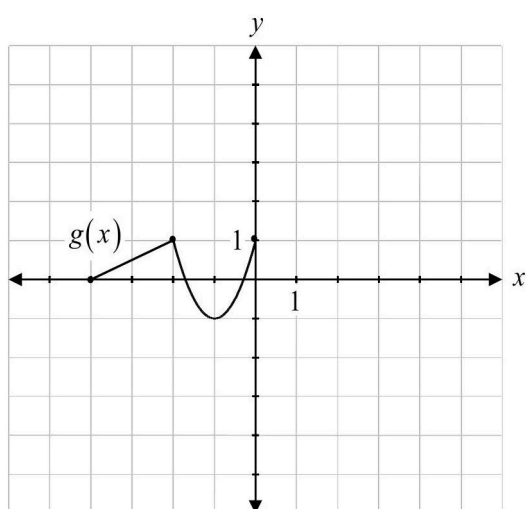
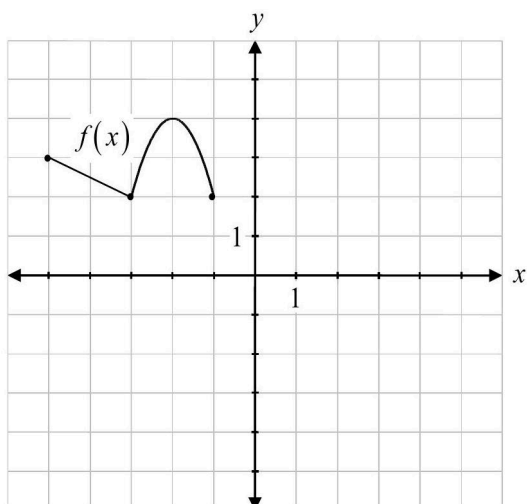
1 mark

or

$$P(x) = x^3 - 3x^2 + 3x + 1$$

Question 13

Determine the equation for $g(x)$ in terms of $f(x)$.



$g(x) =$

—

Solution

$$g(x) = -f(x - 1) + 3$$

1 mark for vertical reflection

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Question 14

Explain why the binomial expansion of $(2x + y)^9$ does not have a middle term.

Solution

The expansion contains $n + 1$ terms. Since n equals 9, there are 10 terms, which would not allow for a middle term.

1 mark

Question 15

Using the laws of logarithms, completely expand the expression $\log\left(\frac{5\sqrt{a}}{b^3}\right)$.

Solution

$$\log 5 + \frac{1}{2} \log a - 3 \log b$$

1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

3 marks

Question 16

Identify 10° in radians.

A.

$$\frac{1800}{\pi}$$

B.

$$\frac{\pi}{1800}$$

C.

$$\frac{18}{\pi}$$

D.

$$\frac{\pi}{18}$$

Solution

Correct answer: D

$$\frac{\pi}{18}$$

Question 17

The polynomial function, $P(x) = a(x - 1)^2(x + 4)^2$, has a y -intercept of -8 .

Identify the value of a .

A.

-2

B.

$-\frac{1}{2}$

C.

$\frac{1}{2}$

D.

2

Solution

Correct answer: B

$-\frac{1}{2}$

Question 18

Identify the value of $\log_4\left(\frac{1}{16}\right)$.

A.

−2

B.

− $\frac{1}{2}$

C.

$\frac{1}{2}$

D.

2

Solution

Correct answer: A

−2

Question 19

Given the angle $\frac{25\pi}{7}$, identify the coterminal angle on the interval $[-2\pi, 0]$.

A.

$$\frac{18\pi}{7}$$

B.

$$\frac{11\pi}{7}$$

C.

$$-\frac{3\pi}{7}$$

D.

$$-\frac{10\pi}{7}$$

Solution

Correct answer: C

$$-\frac{3\pi}{7}$$

Question 20

Identify which expression cannot be evaluated.

A.

$${}_7P_0$$

B.

$${}_7P_6$$

C.

$${}_7P_7$$

D.

$${}_7P_8$$

Solution

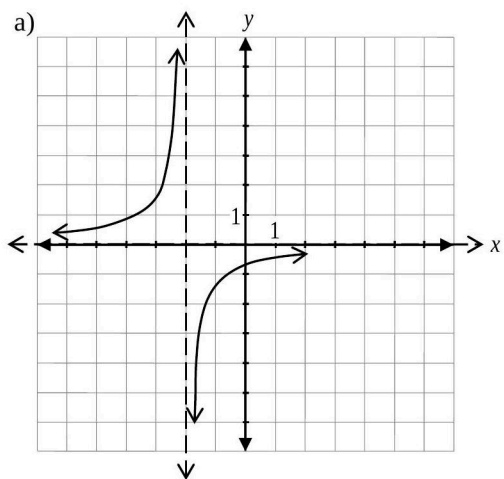
Correct answer: D

$${}_7P_8$$

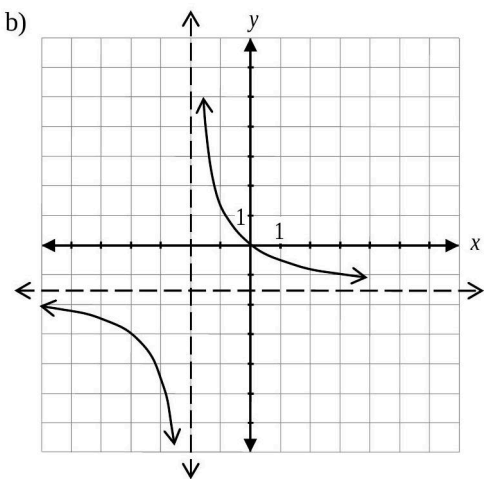
Question 21

Identify the graph of $f(x) = \frac{-3x}{2x+4}$.

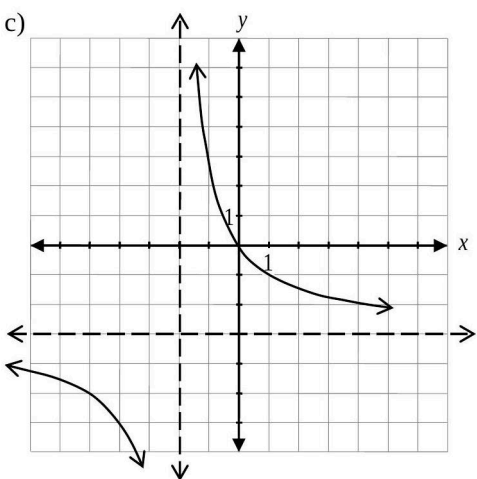
A.



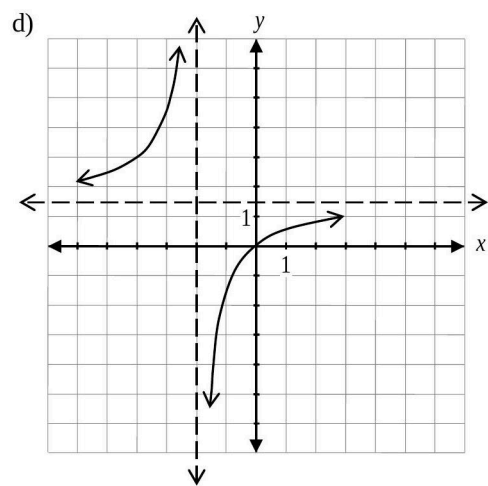
B.



C.

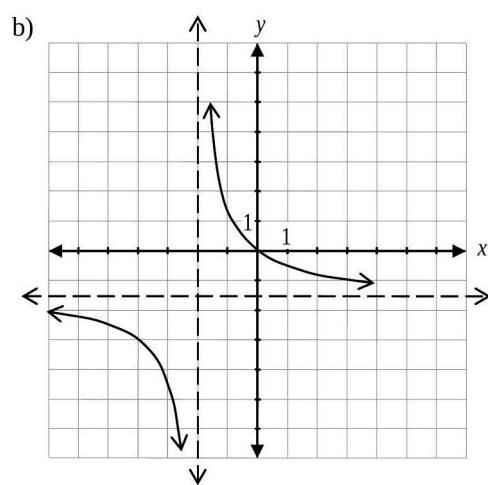


D.



Solution

Correct answer: B



Question 22

Given a point $(-2, 0)$ on the graph of $y = f(x)$, identify the coordinates of the corresponding point on the graph of $y = 4f(\frac{1}{2}x)$.

A.

$(-8, 0)$

B.

$(-4, 0)$

C.

$(-2, 0)$

D.

$(-1, 0)$

Solution

Correct answer: B

$(-4, 0)$

Question 23

Identify the non-permissible value of θ for the expression $\frac{\cos \theta}{1 + \sin \theta}$.

A.

$$\frac{\pi}{2}$$

B.

$$\pi$$

C.

$$\frac{3\pi}{2}$$

D.

$$2\pi$$

Solution

Correct answer: C

$$\frac{3\pi}{2}$$

Question 24

Identify the function with an asymptote at $x = -3$.

A.

$$y = \log(x + 3)$$

B.

$$y = \log x + 3$$

C.

$$y = \log(x - 3)$$

D.

$$y = \log x - 3$$

Solution

Correct answer: A

$$y = \log(x + 3)$$

Question 25

Evaluate the following expression.

$$\tan\left(\frac{2\pi}{3}\right) \csc\left(\frac{-2\pi}{3}\right) + \cos(3\pi)$$

Solution

$$(-\sqrt{3})\left(-\frac{2}{\sqrt{3}}\right) + (-1)$$
$$\frac{2 - 1}{1}$$

1 mark for $\tan\left(\frac{2\pi}{3}\right)$ (1/2 mark for quadrant, 1/2 mark for value)

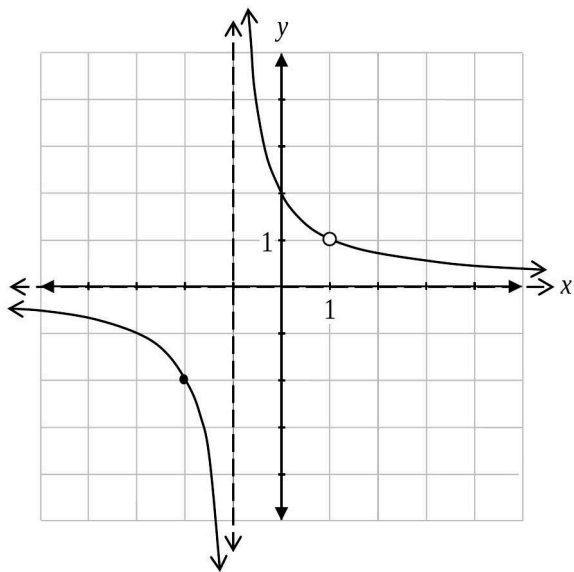
1 mark for $\csc\left(-\frac{2\pi}{3}\right)$ (1/2 mark for quadrant, 1/2 mark for value)

1 mark for $\cos(3\pi)$

3 marks

Question 26

State the range of the graph below.



Range:

Solution

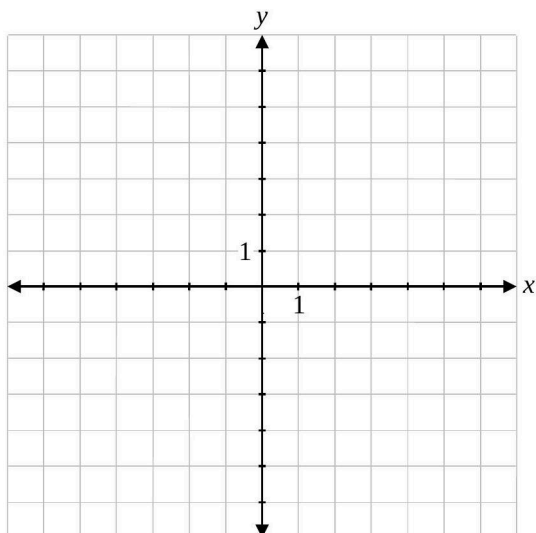
$\{y \in \mathbb{R}, y \neq 0 \text{ and } y \neq 1\}$

1 mark ($\frac{1}{2}$ mark for $y \neq 0$, $\frac{1}{2}$ mark for $y \neq 1$)

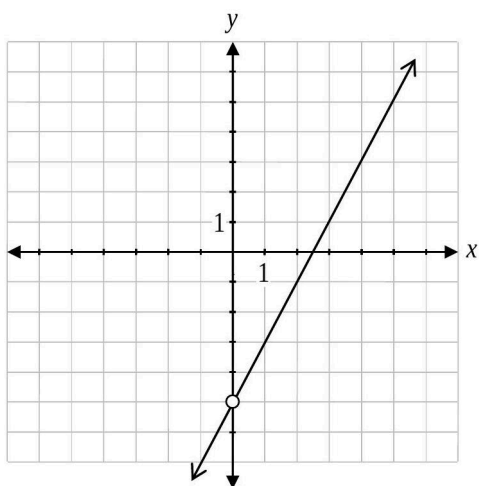
1 mark

Question 27

Sketch the graph of the function $f(x) = \frac{2x^2 - 5x}{x}$.



Solution



1 mark for point of discontinuity (hole) at $(0, -5)$

$(\frac{1}{2}$ mark for $x = 0$, $\frac{1}{2}$ mark for $y = -5$)

1 mark for shape of a linear function

2 marks

Question 28

State a possible value of n if the polynomial function $P(x) = (x - 1)^2(x + 2)^n$ has a range of $[0, \infty)$.

Solution

$$n = 2$$

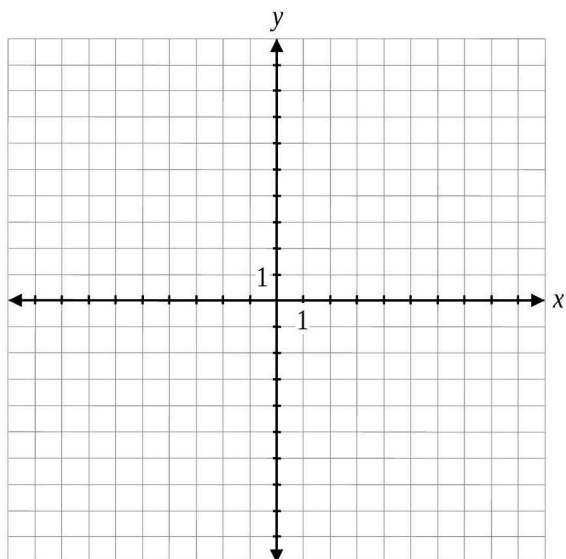
1 mark

Note(s):

- Accept any even positive value of n , including zero.

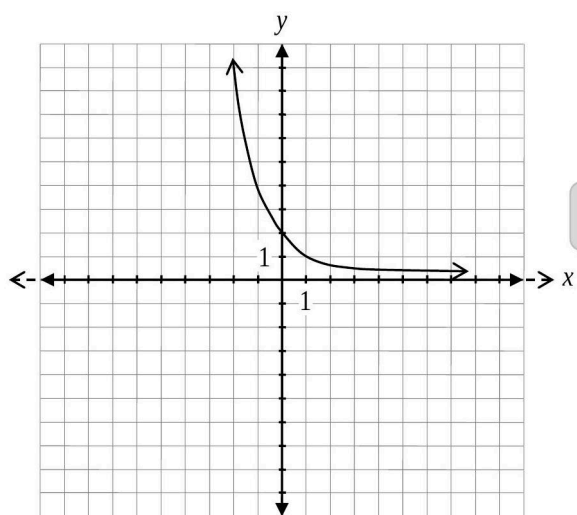
Question 29

Sketch the graph of $y = \left(\frac{1}{2}\right)^{x-1}$.



Solution

Solution



1 mark for decreasing exponential function

1 mark for horizontal translation

2 marks

Question 30

Solve.

$$\log_x 27 = 3$$

Solution

$$x^3 = 27$$

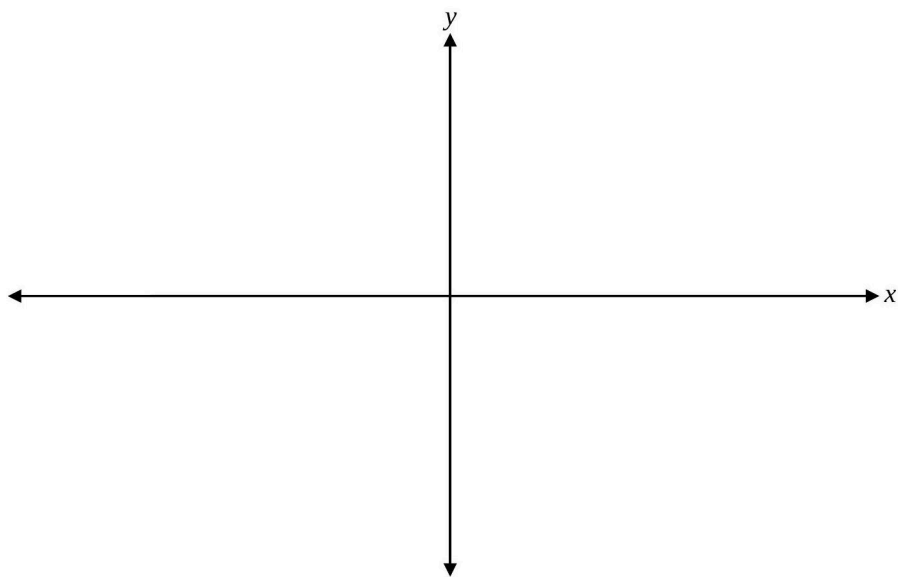
$$x = 3$$

1 mark for exponential form

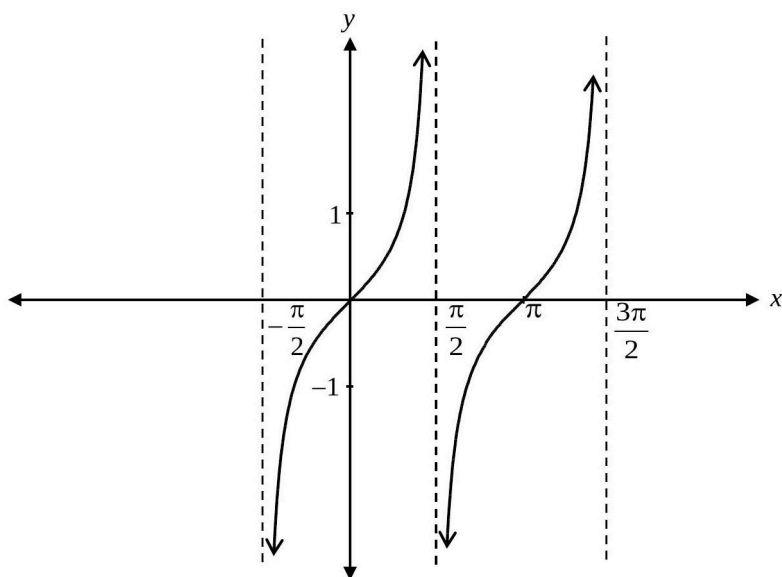
1 mark

Question 31

Sketch at least two periods of the graph $y = \tan x$.



Solution



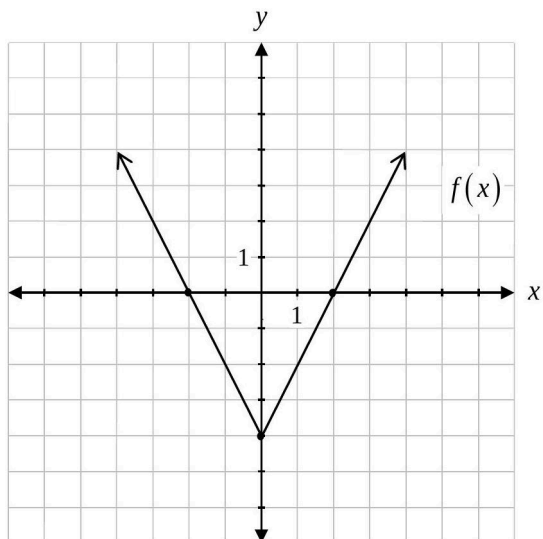
1 mark for increasing trigonometric function

1 mark for asymptotic behaviour approaching $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

2 marks

Question 32

Given the graph of $f(x)$, determine the domain of $\frac{1}{f(x)}$.



Domain:

Solution

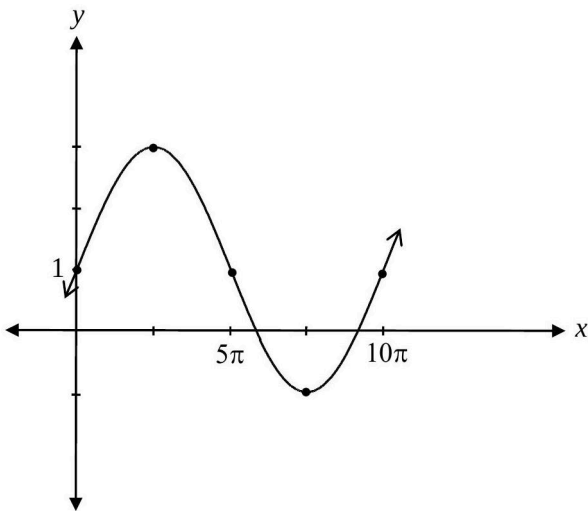
Domain: $\{x \in \mathbb{R}, x \neq \pm 2\}$

1 mark ($\frac{1}{2}$ mark for $x \neq 2$, $\frac{1}{2}$ mark for $x \neq -2$)

1 mark

Question 33

Determine the values of A, B, and D of the sinusoidal function in the form $y = A \sin(Bx) + D$.



A =

B =

D =

Solution

$A = 2$

$B = \frac{1}{5}$

1 mark for B

$D = 1$

1 mark for D

□

Question 34

Determine if the point $\left(-\frac{\sqrt{7}}{5}, \frac{2}{5}\right)$ is on the unit circle.

Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-hand side} &= \left(-\frac{\sqrt{7}}{5}\right)^2 + \left(\frac{2}{5}\right)^2 \\ &= \frac{7}{25} + \frac{4}{25} \\ &= \frac{11}{25}\end{aligned}$$

$$\frac{11}{25} \neq 1$$

$\therefore$ not on the unit circle

1 mark for justification

1 mark

Question 35

Solve, algebraically.

$$\frac{{}_nC_5}{{}_nC_4} = 6$$

Solution

$$\frac{\frac{n!}{(n-5)!5!}}{\frac{n!}{(n-4)!4!}} = 6$$

$$\frac{n!(n-4)(n-5)!4!}{n!(n-5)!5 \cdot 4!} = 6$$

$$\frac{n-4}{5} = 6$$

$$n-4 = 30$$

$$n = 34$$

$\frac{1}{2}$ mark for substitution into equation

1 mark for factorial expansion

($\frac{1}{2}$ mark for numerical factors; $\frac{1}{2}$ mark for factors with variables)

1 mark for simplification of factorials

($\frac{1}{2}$ mark for numerical factors; $\frac{1}{2}$ mark for factors with variables)

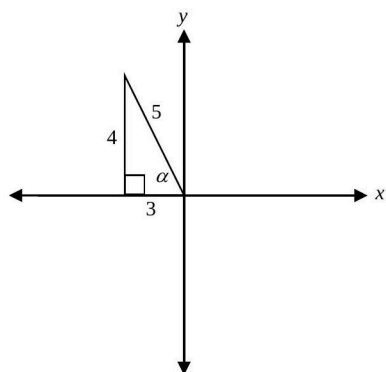
$\frac{1}{2}$ mark for solving for n

3 marks

Question 36

Given $\sin \alpha = \frac{4}{5}$, where α is in quadrant II, determine the exact value of $\sin 2\alpha$.

Solution



$$x^2 + y^2 = r^2$$

$$x^2 + 16 = 25$$

$$x^2 = 9$$

$$x = \pm 3$$

$$x = -3$$

$$\cos \alpha = -\frac{3}{5}$$

1/2 mark for value of x

1/2 mark for $\cos \alpha$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$= 2 \left(\frac{4}{5} \right) \left(-\frac{3}{5} \right)$$

$$= -\frac{24}{25}$$

1 mark for substitution into correct identity

2 marks

- Accept any of the following values for x : $x = \pm 3$; $x = -3$; or $x = 3$.

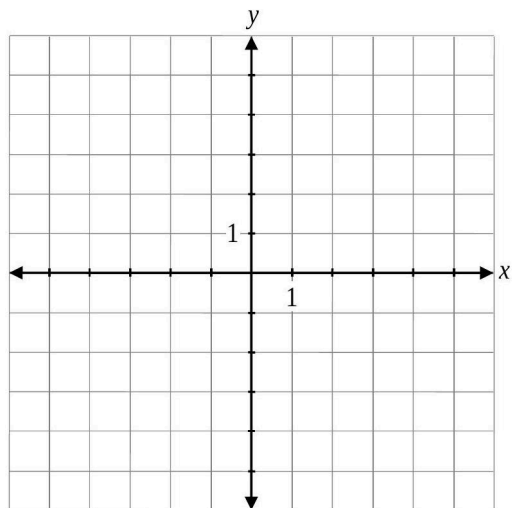
Question 37

Given the functions $f(x) = x + 1$ and $g(x) = \sqrt{x}$,

a) determine the equation of $g(f(x))$.

$$g(f(x)) =$$

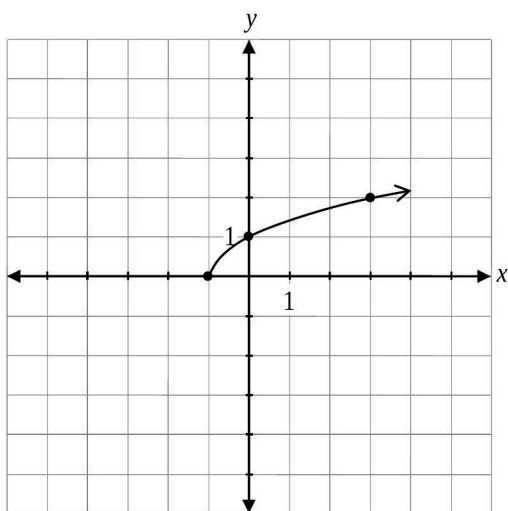
b) sketch the graph of $g(f(x))$.



Solution

a) $g(f(x)) = \sqrt{x+1}$

b)



1 mark

1 mark for domain of $g(f(x))$

1 mark for shape consistent with $g(f(x))$

2 marks

Question 38

Steve is asked to determine an equation with a larger period than the period of the graph of

$$y = \cos(2x).$$

Justify why Steve's answer of $y = \cos(6x)$ is incorrect.

Solution

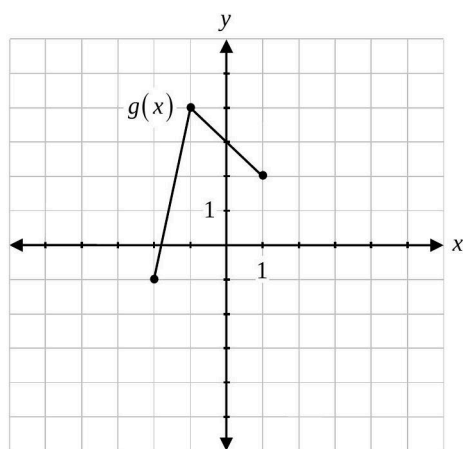
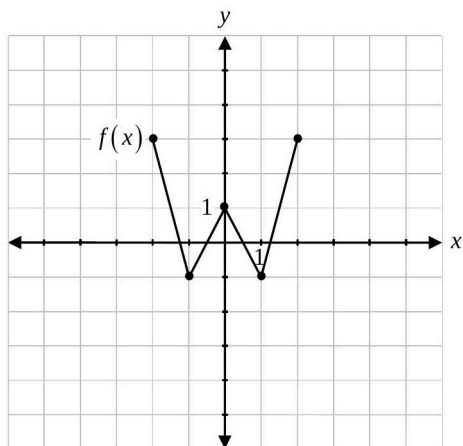
Steve's equation needs to have a value of $|b|$ less than 2. 1 mark

or

Steve's graph would have a period of $\frac{2\pi}{6} = \frac{\pi}{3}$, which is smaller than $\frac{2\pi}{2} = \pi$, the period of the given graph.

Question 39

Given the graphs of $f(x)$ and $g(x)$,



- determine the value of $(f \cdot g)(-1)$.
- determine the value of $g(f(0))$.

Solution

a)

$$\begin{aligned}(f \cdot g)(-1) &= (-1)(4) \\ &= -4\end{aligned}$$

b) $f(0) = 1$

$$g(f(0)) = 2$$

1 mark for value of $(f \cdot g)(-1)$

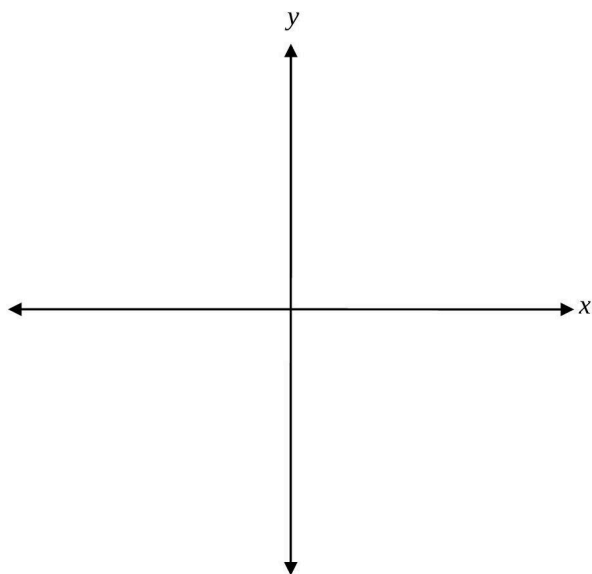
1 mark

$\frac{1}{2}$ mark for $f(0)$

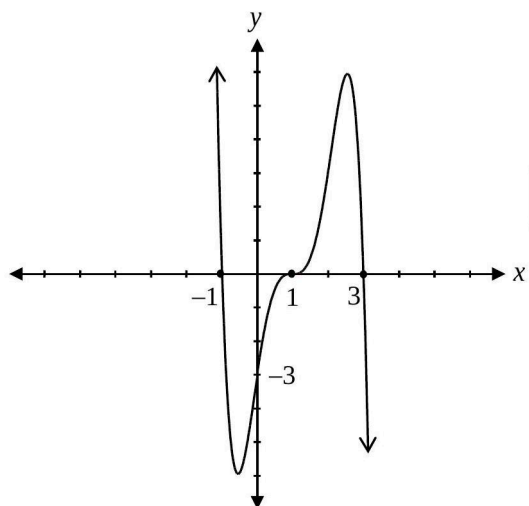
$\frac{1}{2}$ mark for $g(f(0))$ consistent with $f(0)$ value

Question 40

Sketch the graph of $P(x) = -(x-1)^3(x-3)(x+1)$.



Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity (degree 3 at $x = 1$)

$\frac{1}{2}$ mark for end behaviour

3 marks

Question 41

The point $(-\sqrt{3}, 1)$ is on the terminal arm of an angle θ , in standard position.

- a) Determine $\tan \theta$.
- b) Determine a possible value of θ , in radians.

Solution

a) $\tan \theta = -\frac{1}{\sqrt{3}}$

b) $\theta = \frac{5\pi}{6}$

Question 42

Describe the transformation used to obtain the graph of $y = \log_5 x$ given the graph of $y = 5^x$.

Solution

The graph of $y = \log_5 x$ is obtained by reflecting the graph of $y = 5^x$ over the line $y = x$.

or

1 mark

The graph of $y = \log_5 x$ is the inverse of $y = 5^x$.

Question 43

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$, where $\theta \in \mathbb{R}$.

Solution

$\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$ 1 mark for values of θ ($\frac{1}{2}$ mark for each value)

$\theta = \begin{cases} \frac{4\pi}{3} + 2k\pi, k \in \mathbb{Z} \\ \frac{5\pi}{3} + 2k\pi, k \in \mathbb{Z} \end{cases}$ 1 mark for general solution

or

$$\theta = 240^\circ, 300^\circ$$

$$\theta = \begin{cases} 240^\circ + 360^\circ k, k \in \mathbb{Z} \\ 300^\circ + 360^\circ k, k \in \mathbb{Z} \end{cases}$$

Question 44

Given that the point (a, b) is on the graph of $f(x)$, describe how you would determine the corresponding point on the graph of $y = \sqrt{f(x)}$.

Solution

The value of a stays the same, square root the value of b .

1 mark

Question 45

Evaluate.

$$\cos\left(\frac{\pi}{20}\right)\cos\left(\frac{\pi}{5}\right) - \sin\left(\frac{\pi}{20}\right)\sin\left(\frac{\pi}{5}\right)$$

Solution

$$\cos\left(\frac{\pi}{20} + \frac{\pi}{5}\right)$$

$$\cos\left(\frac{\pi}{20} + \frac{4\pi}{20}\right)$$

$$\cos\left(\frac{5\pi}{20}\right)$$

$$\cos\left(\frac{\pi}{4}\right)$$

$$\frac{\sqrt{2}}{2}$$

$\frac{1}{2}$ mark for substitution of an appropriate identity

$\frac{1}{2}$ mark for exact value

1 mark

Question 46

Describe the transformations used to obtain the graph of the function $y = f(-x + 6) - 8$ from the graph of $y = f(x)$.

Solution

Reflect the graph of $y = f(x)$ over the y -axis and then translate 6 units right and 8 units down.

1 mark for horizontal reflection

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Note(s):

- Deduct 1 mark if correct transformations are given in the wrong order.

Question 47

State the equations of all the asymptotes of the function, $y = \frac{1}{3x+1}$.

Solution

$$y = 0$$

$$x = -\frac{1}{3}$$

1 mark for horizontal asymptote

1 mark for vertical asymptote

2 marks

Question 48

Determine the zeros of the polynomial function $P(x) = 2x^3 + 5x^2 - 4x - 3$.

Solution

$P(1) = 2(1)^3 + 5(1)^2 - 4(1) - 3$ 1 mark for identifying one possible value of x

$P(1) = 0$

$(x - 1)$ is a factor

1	2	5	-4	-3
	↓	2	7	3
	2	7	3	0

1 mark for synthetic division (or for any equivalent strategy)

$$P(x) = (x - 1)(2x^2 + 7x + 3)$$
$$0 = (x - 1)(2x + 1)(x + 3)$$
$$x = 1 \quad x = -\frac{1}{2} \quad x = -3$$

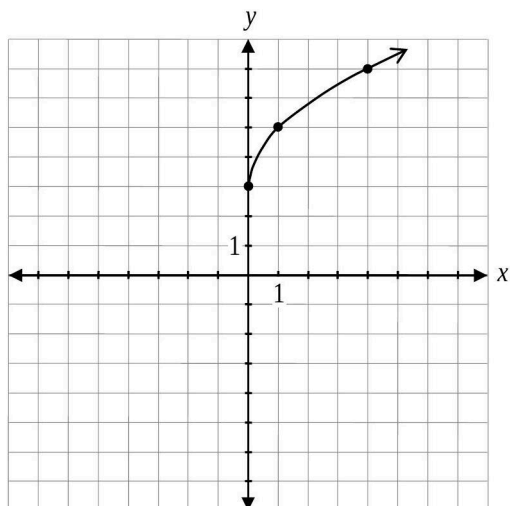
$\frac{1}{2}$ mark for consistent factors

$\frac{1}{2}$ mark for all zeros

3 marks

Question 49

Determine the equation of the radical function represented by the graph.



$y =$

Solution

$$y = 2\sqrt{x} + 3 \quad 1 \text{ mark for vertical stretch}$$

1 mark for vertical translation

$y =$ or

$$\sqrt{4x} + 3$$

1 mark for horizontal stretch

1 mark for vertical translation

2 marks